Assignment 02: DA5402

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REPORT

In this assignment, the main goal was to manage a repository with two developers working on it simultaneously. Each developer had to develop 2 features each and serve them on a common endpoint.

Pranav:

Features : NER and Speech Synthesis
NER: Used Spacy to do so
TTS: Used pyt2s to do so

Sanved:

Features : Translate and Image Generation
Translate: Used deep-translate, which is a wrapper for multiple translation tools
Image Gen: Used HF interface for Stability AI

No of branches : 4
No of Pull requests raised : 11
No of Conflicts resolved : 3

Issues faced:

Pranav :

Dependency Hell - Tried to use Huggingface model, but there were multiple dependency issues while building between packages as well cyclic import problems; Decided to find a simple open-source API endpoint for TTS

Appreciate the services being served as a single bundle with all dependencies

Sanved:

Faced issues with different img generation APIs.

Tried to run a simple model for translation locally. Ran into computational infeasibility.

Shifted to a python wrapper for existing APIs.

PART 2

Issues faced and how they were resolved:

1. The app initially worked fine when run locally. It didn't create any issues. When Dockerized, the app had multiple issues and had to be refactored multiple times for it to work inside the Docker image.
2. Docker had to be given the variables inside `.env`. This was done. The `.env` file was mentioned in the `.dockerignore` file, so it wasn't copied inside the machine. It was passed in separately.
3. Although I used CLI Docker, my Docker engine was run by Docker Desktop. Docker Desktop manages a single Docker node in a VM. It doesn't work with Swarm. So I removed all instances of Docker from the system.

   I `apt install`ed Docker once more, without Docker Desktop. This time, the Swarm mode started, but the worker node still was unable to connect.

4. The worker node still wasn't able to connect. So I passed the specific address I wanted Docker Swarm to run on. This was one of the free IPs i got from `hostname -I`.

AI Disclosure Appendix

Pranav:
AI (Gemini 3 Pro) was used as a search engine for debugging and getting information about present API endpoints. The files are 100% human-generated (by myself) in reference to the files in Assignment 1 (Even the HTML file this time is written by me).
This holds for part2 as well, these files are 100% human-generated as well.

Sanved:
I have not used AI to write any code. Every single code file was written by me.
Google search AI was used to go through API documentation. No code was copied from AI models.
I have referred to code examples from API documentation, they were not used directly, but with heavy modification.
For part 2:
I used AI to understand how Docker swarm works, to clear a few doubts.
No code from AI was reproduced in the project/assignment.
AI was used to debug errors that i got and search up a few linux commands for me to debug stuff on my machine, since it was the manager and got quite a few errors at first.